```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('Customer Churn.csv')
df.head()
```

Show hidden output

```
#replacing the blanks with zero
df["TotalCharges"] = df["TotalCharges"].replace(" ","0")
df["TotalCharges"] = df["TotalCharges"].astype("float")
df.info()
```

Show hidden output

```
df.describe()
```

Show hidden output

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

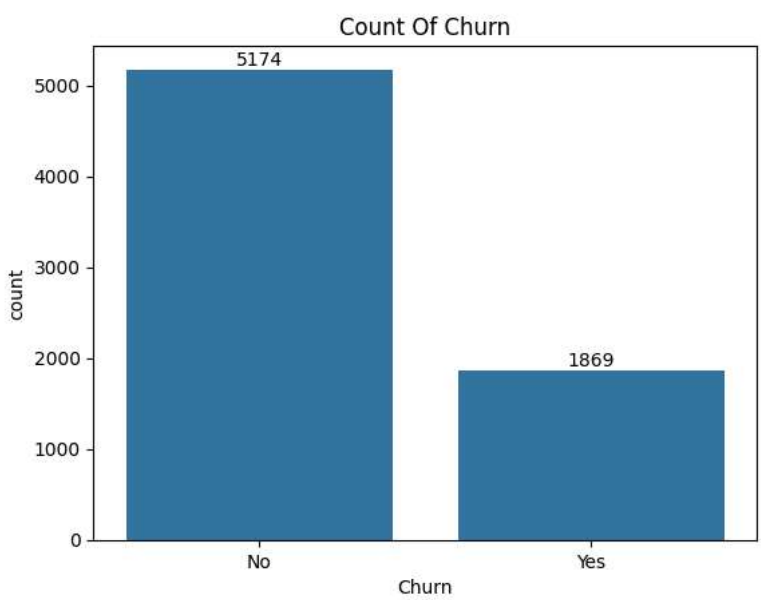
```
df["customerID"].duplicated().sum()
```

```
np.int64(0)
```

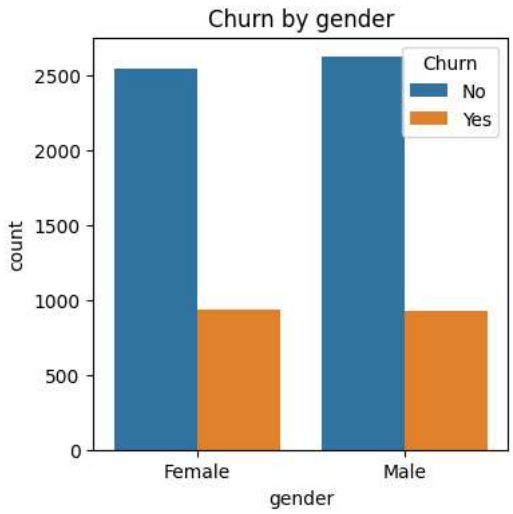
```
def conv(value):
    if value == 1:
        return "yes"
    else:
        return "no"

df["SeniorCitizen"] = df["SeniorCitizen"].apply(conv)
df.head(22)
```

Show hidden output

```
ax=sns.countplot(x ='Churn' , data = df)
ax.bar_label(ax.containers[0])
plt.title('Count Of Churn')
plt.show()
```



```
plt.figure(figsize=(4,4))
sns.countplot(data=df,x='gender',hue='Churn')
plt.title('Churn by gender')
plt.show()
```




```
plt.figure(figsize=(4,4))
ax = sns.countplot(data=df, x='SeniorCitizen', hue='Churn')
plt.title('Churn by SeniorCitizen')

for container in ax.containers:
    ax.bar_label(container)

plt.show()
```

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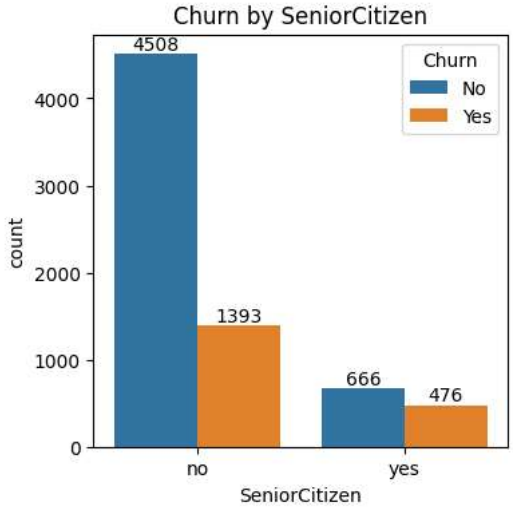
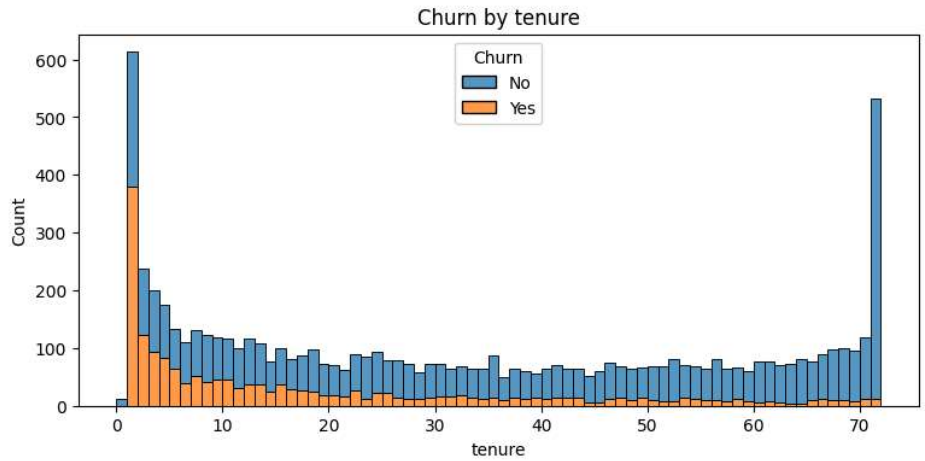
comparatively a greater percentage of people in senior citizen category have churned out




```
plt.figure(figsize=(9,4))
sns.histplot(data=df,x='tenure',hue='Churn',bins=72,multiple='stack')
plt.title('Churn by tenure')
plt.show()
```



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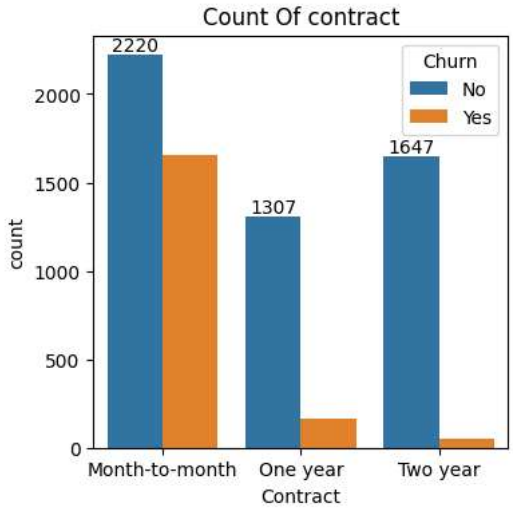
customers having month to month contract churn out more often compared to one year or two years

```
plt.figure(figsize=(4,4))
ax=sns.countplot(data=df,x='Contract',hue='Churn')
ax.bar_label(ax.containers[0])
plt.title('Count Of contract')
plt.show()
```




```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLi
0	7590-VHVEG	Female	no	Yes	No	1	No	No ph ser
1	5575-GNVDE	Male	no	No	No	34	Yes	
2	3668-QPYBK	Male	no	No	No	2	Yes	
3	7795-CFOCW	Male	no	No	No	45	No	No ph ser
4	9237-HQITU	Female	no	No	No	2	Yes	

5 rows × 21 columns

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The plots show that churn is higher among customers with month-to-month contracts, paperless billing, and electronic payment methods.
Customers with fiber optic internet, no online security, and no tech support also tend to churn more.
On the other hand, longer-term contracts and added support services are linked with lower churn rates.

```
import matplotlib.pyplot as plt
import seaborn as sns

# Categorical columns
categorical_cols = ['gender', 'SeniorCitizen', 'Partner', 'Dependents',
                    'PhoneService', 'MultipleLines', 'InternetService',
```

```
                        'PhoneService', 'MultipleLines', 'InternetService',
                        'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
                        'TechSupport', 'StreamingTV', 'StreamingMovies',
                        'Contract', 'PaperlessBilling', 'PaymentMethod']

# Subplot grid
n_cols = 3
n_rows = (len(categorical_cols) + n_cols - 1) // n_cols

plt.figure(figsize=(20, n_rows * 5))

for idx, col in enumerate(categorical_cols, 1):
    plt.subplot(n_rows, n_cols, idx)
    sns.countplot(data=df, x=col, hue="Churn", palette=["skyblue", "orange"])
    plt.title(f'Count Plot of {col} by Churn')
    plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

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customer is more likely to churn when he is using electronic churn

```
df.columns
```

```
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
       'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
       'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
```

```
from typing import Container

sns.countplot(data=df,x='PaymentMethod',hue='Churn')
plt.title('churned customers by Payment Method')
ax=sns.countplot(data=df,x='PaymentMethod',hue='Churn')
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.xticks(rotation =10)
plt.show()
```